BALL — foundation-model performance

Main classifiers — Accuracy

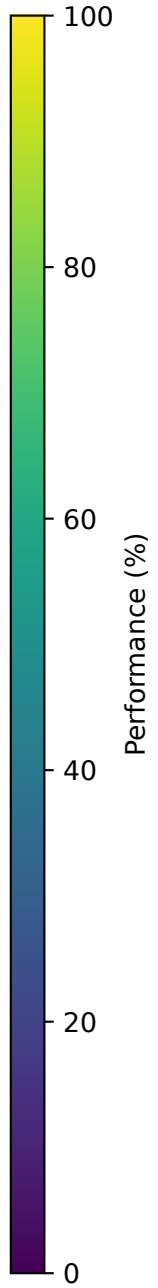
	Omiclass	Logistic Regression
DE baseline		
DE	97.6 ±0.5	N/A
BulkFormer		
Full	70.9	82.8
PCA	77.6	82.2
Supervised	68.8	81.3
Supervised + PCA	78.6	79.2
CellFM 1-view		
Full	14.3	43.6
PCA	18.4	26.6
Supervised	10.9	24.5
Supervised + PCA	18.2	16.4
CellFM 20-view		
Full	88.1	96.0
PCA	93.5	95.0
Supervised	84.1	86.0
Supervised + PCA	87.0	88.7
BulkRNABert		
Full	53.9	76.1
PCA	62.5	71.7
Supervised	54.3	73.8
Supervised + PCA	60.4	70.9

Main classifiers — Macro-F1

	Omiclass	Logistic Regression
DE	95.0 ±1.2	N/A
BulkFormer		
Full	47.2	64.9
PCA	60.9	66.6
Supervised	42.9	62.1
Supervised + PCA	63.9	65.4
CellFM 1-view		
Full	4.5	29.2
PCA	7.3	17.1
Supervised	2.9	17.1
Supervised + PCA	9.6	10.6
CellFM 20-view		
Full	77.7	93.1
PCA	88.0	91.3
Supervised	72.7	72.8
Supervised + PCA	77.4	78.5
BulkRNABert		
Full	40.2	61.0
PCA	46.4	54.6
Supervised	35.8	59.5
Supervised + PCA	45.9	54.1

Additional classifiers — Accuracy

	MLP	MLP + SMOTE	SVM
BulkFormer			
Full	78.0	74.2	59.5
PCA	79.9	79.2	65.8
Supervised	76.9	72.3	65.6
Supervised + PCA	78.0	79.0	66.7
Full tokens	92.9	N/A	N/A
CellFM 1-view			
Full	22.2	19.3	13.0
PCA	26.8	25.4	18.9
Supervised	21.6	22.0	15.9
Supervised + PCA	19.7	18.2	17.6
Full tokens	14.9	N/A	N/A
CellFM 20-view			
Full	81.6	58.9	86.6
PCA	94.3	94.3	93.1
Supervised	83.4	80.3	82.4
Supervised + PCA	87.4	87.0	84.7
Full tokens	84.7	N/A	N/A
BulkRNABert			
Full	63.5	50.3	37.5
PCA	64.2	65.2	60.4
Supervised	63.3	51.2	41.5
Supervised + PCA	61.0	62.7	61.2

Additional classifiers — Macro-F1

	MLP	MLP + SMOTE	SVM
BulkFormer			
Full	56.8	60.4	33.3
PCA	63.8	66.6	53.4
Supervised	59.7	55.5	42.7
Supervised + PCA	61.6	61.3	53.9
Full tokens	84.4	N/A	N/A
CellFM 1-view			
Full	13.7	11.7	3.5
PCA	14.5	13.9	6.4
Supervised	13.4	12.5	4.6
Supervised + PCA	10.8	10.9	6.2
Full tokens	5.8	N/A	N/A
CellFM 20-view			
Full	61.3	44.0	71.8
PCA	88.0	88.7	88.5
Supervised	75.0	66.6	72.0
Supervised + PCA	76.6	75.5	74.8
Full tokens	62.7	N/A	N/A
BulkRNABert			
Full	42.4	38.9	22.3
PCA	50.9	49.6	44.5
Supervised	44.8	40.0	25.3
Supervised + PCA	45.6	50.5	47.1